

Session Objectives

1. Describe the different types of fungal and parasitic CNS infections and their etiology
2. Compare and contrast parasitic and fungal meningitis
3. Compare the properties of the parasitic and fungal etiological agents of meningitis, encephalitis, myelitis and brain abscesses
4. Identify the importance of diagnostic CSF analysis and antimicrobial therapy
5. Illustrate epidemiological factors in the determination of etiology

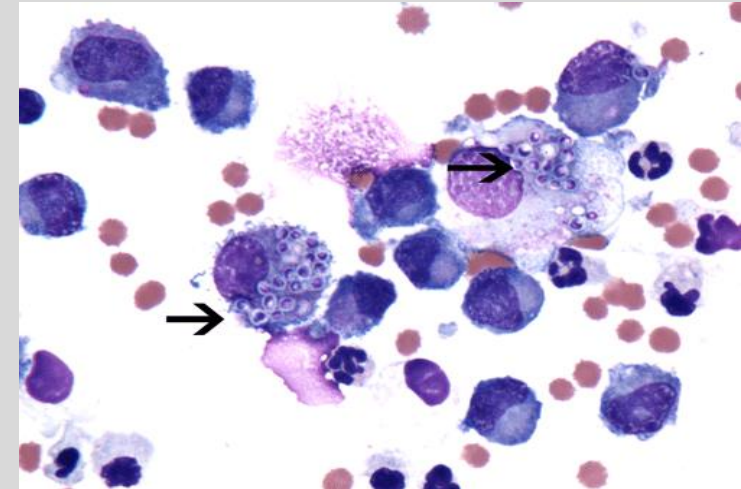
Source: Course Syllabus

CNS Infection (Chronic Meningitis)

- Chronic Meningitis
 - Meningeal inflammation with CSF pleocytosis > 4 weeks
 - Symptoms evolve over days/weeks
 - a) Headache, fever, etc.
 - b) Nuchal rigidity subtle/absent
 - c) Progresses $\rightarrow$ seizures, mental status changes, confusion or hallucinations
 - d) Focal neurologic deficits develop
 - e) Hydrocephalus & increased intracranial pressure

Fungal Chronic Meningitis

- *Histoplasma capsulatum*
 - Rare
 - Endemic regions – the Ohio River Valley
 - Presentation: fever, oral lesions, hepatosplenomegaly, hydrocephalus
 - Laboratory:
 - a) Culture (27-65% of cases)
 - b) *Histoplasma* polysaccharide antigen in the urine/blood/CSF (61% of patients)
 - c) Visualization of yeast cells inside macrophages
 - Treatment: amphotericin B



Fungal Chronic Meningitis

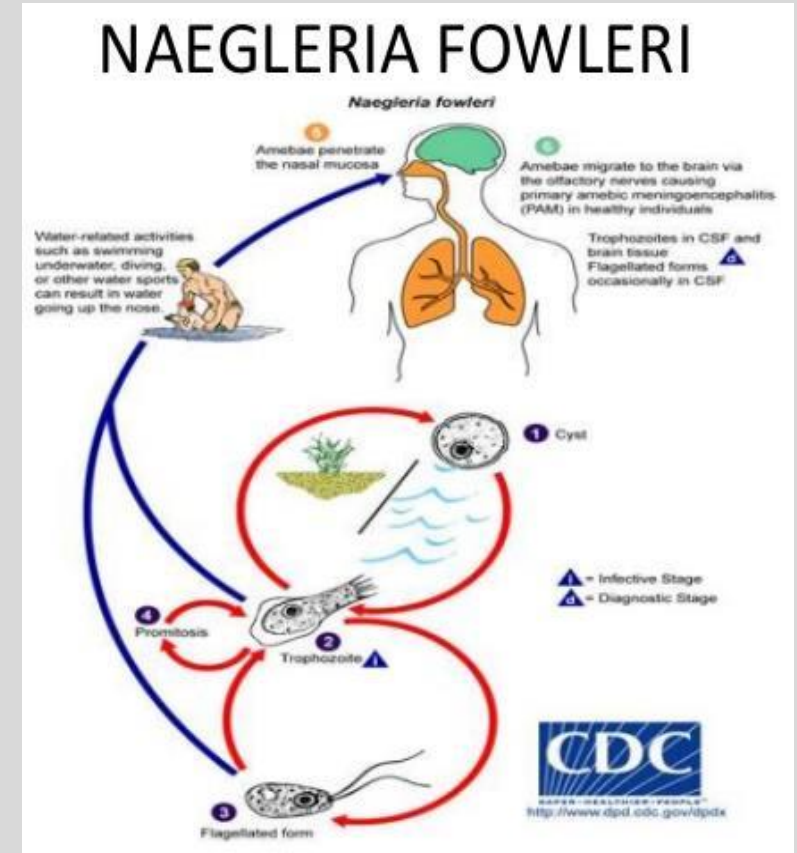
- *Pseudallescheria* spp. (*P. boydii*)
- Most common fungal etiologic agent of mycetoma
- Meningitis, meningoencephalitis & brain abscesses
- Most in immunocompromised (but seen in immunocompetent)
- Polluted water, sewage, swamps
- Near drowning victims
- Resistant to Amphotericin B (miconazole shows promise)

CNS Infection (Encephalitis)

- Inflammatory process in the brain
- Associated with clinical evidence of neurologic dysfunction
- Most challenging syndromes to diagnose
- Specific pathogen is identified in $< 50\%$ of cases
 - a) Vast number of infectious causes
 - b) Many noninfectious etiologies that present with similar syndromes
 - c) Confusion regarding the role of molecular versus serologic testing
- Parasitic: less commonly identified, but are neurotropic & cause sporadic encephalitis and myelitis

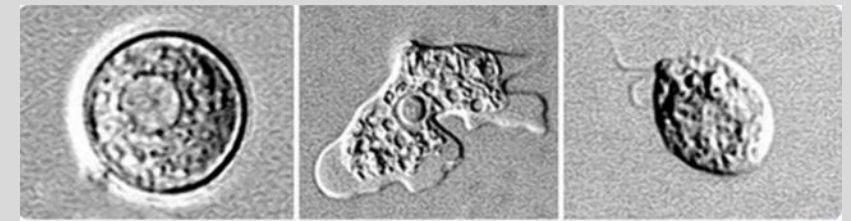
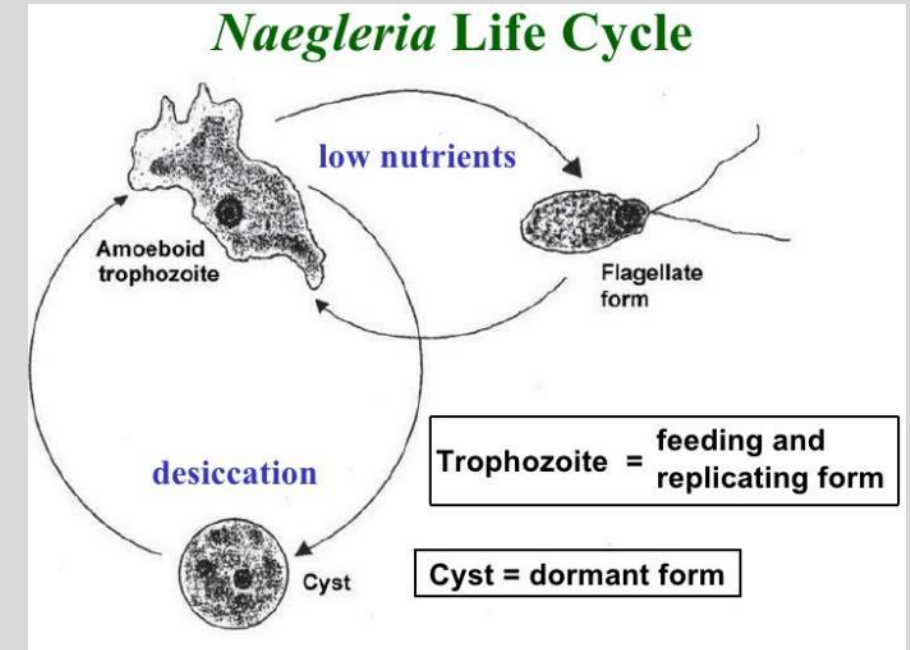
Parasitic encephalitis

- *Naegleria fowleri*
- Protozoan
- Amebic meningoencephalitis (PAM; Brain-eating disease)
- 100% mortality
- Entry: nose while swimming (not drinking)
- In US: freshwater located in southern-tier states
- Thermophile (46°C)



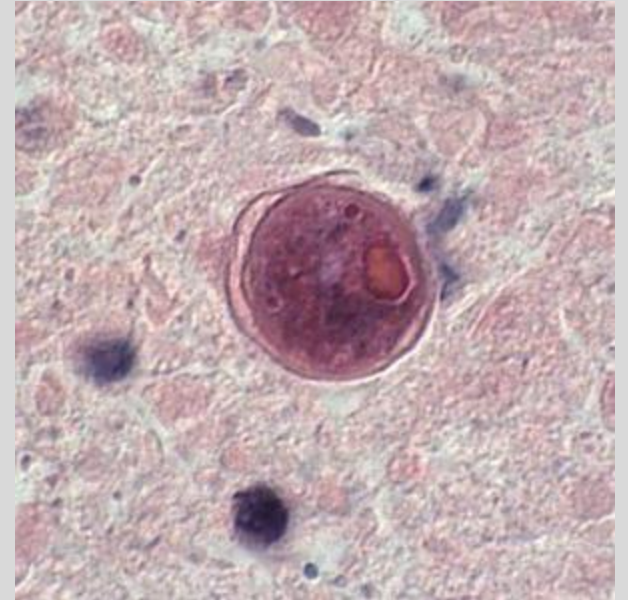
Parasitic encephalitis (*N. fowleri*)

- Symptoms: altered level of consciousness, loss of the sense of smell
- Laboratory:
 - a) Neutrophilic pleocytosis
 - b) Motile trophozoites on wet mount of warm CSF
 - c) Brain histopathology
- Epidemiology:
 - a) Summer months
 - b) Children & adolescent boys at highest risk
 - c) Swimming in fresh water, & particularly water sports
- Treatment:
 - a) High mortality even with treatment
 - b) Miltefosine shows promise



Parasitic encephalitis (*B. mandrillaris*)

- Mortality and Prognosis:
 - a) Poor prognosis
 - b) Slow progressive disease (2 years)
 - c) 89% mortality rate
- Laboratory:
 - a) Lymphocyte pleocytosis and depressed glucose (confused with TB)
 - b) Serology (Indirect immunofluorescence antibody)
 - c) Brain histopathology
 - d) Real-time PCR of brain tissue
- Treatment:
 - a) Sulfadiazine, a macrolide antibiotic (azithromycin or clarithromycin), rifampin, & fluconazole



Cyst of *B. mandrillaris* in brain tissue

Brain abscesses

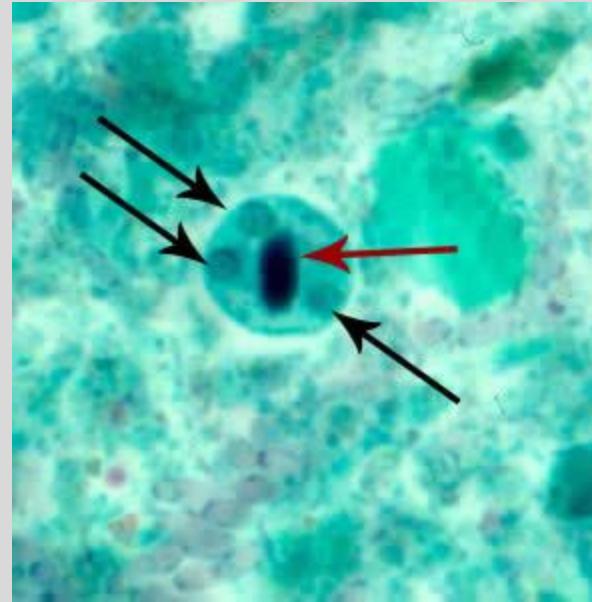
- 2-3 x more common in males than females
- More common in young children and adults over 60
- Predisposing condition: Otitis media or mastoiditis
- Mostly bacterial in etiology
- Parasites that can cause brain abscess:
 - a) *Taenia solium* (cysticercosis)*Entamoeba histolytica*
 - b) *Schistosoma japonicum*
 - c) *Paragonimus* spp.
- Fungi when immunocompromised
 - a) *Aspergillus* spp. are especially common in patients who have bone marrow & transplants
 - b) T-cell immunity defects (including AIDS patients) are predisposed to *Cryptococcus neoformans*
 - c) Neutropenia patients predisposed to *Aspergillus*, *Mucor*, & *Candida*

Brain abscesses

- Clinical manifestations are largely due to presence of space-occupying lesions
- Most common symptoms:
 - a) Headache (75% of patients)
 - b) Altered mental status
 - c) Focal neurologic findings (especially hemiparesis) (>60%)
 - d) Fever (50%)
 - e) Nausea & vomiting (25–50%)
 - f) Seizures (generalized; 30%)
 - g) Other signs/symptoms depending on the abscess stage, size & anatomic location
- Labs:
 - a) Peripheral leukocytosis (40% normal)
 - b) Low serum Na^+ (ADH)
 - c) Platelet counts may be high or low
 - d) Elevated erythrocyte sedimentation rate
 - e) Elevated CRP

Entamoeba histolytica

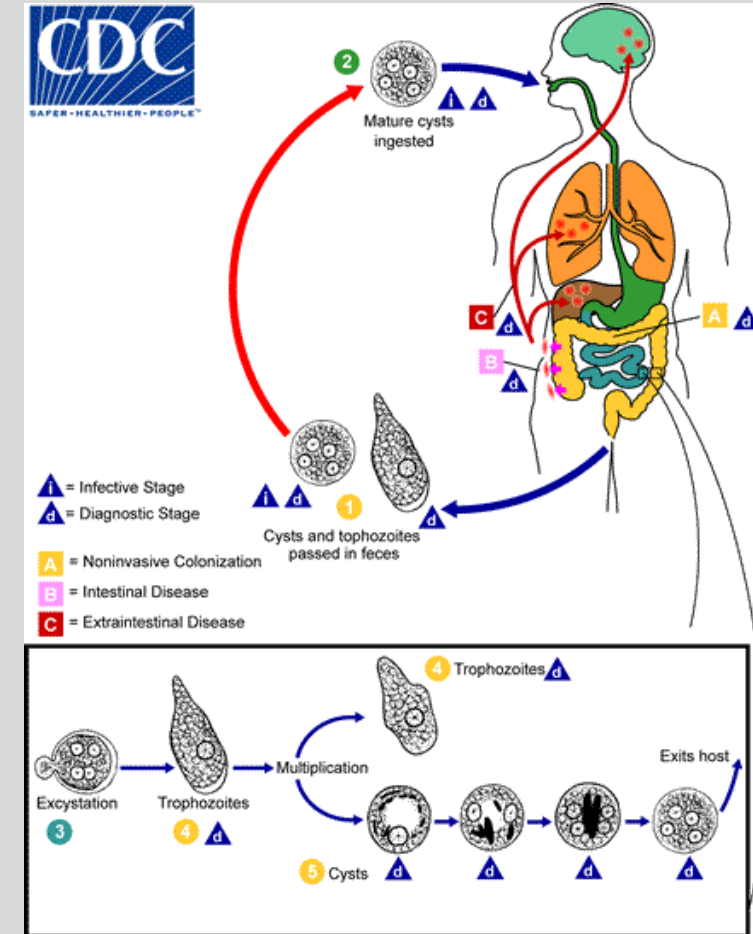
- Predominantly a GI disease
- Diagnosis: serology & PCR of brain abscess aspirate or CSF
- Treatment:
 - Metronidazole or tinidazole
 - Followed by paromomycin or iodoquinol



Cyst stained with trichrome. 3 nuclei are visible (black arrows), and the cyst contains a chromatoid body with typically blunted ends (red arrow)



Trophozoite with ingested erythrocytes



Summary Slide

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